

Mithun Karthikeyan

mithunkar2004@gmail.com | linkedin.com/in/mithunkarth | github.com/mithunkar | Portland, OR

Education

Oregon State University

M.S. in Artificial Intelligence

Corvallis, OR

Sept. 2025 - June 2026

Oregon State University

B.S. in Computer Science, Data Science Option; GPA: 4.0

Corvallis, OR

Sept. 2022 - June 2025

Technical Skills

Languages: Python, C, C++, Java, JavaScript/TypeScript, SQL, R, HTML/CSS

AI and Data: PyTorch, Scikit-learn, NumPy, SciPy, Pandas, ChromaDB, RAG, ETL pipelines

Product and Cloud: React, Node.js, FastAPI, Flask, AWS S3/EC2/DynamoDB, BeautifulSoup, boto3

Systems and Research: Linux/Unix, Git, CI workflows, automated testing, HPC, OpenMP, CUDA, NMR software

Experience

Graduate Teaching Assistant

Oregon State University

Sept. 2025 - June 2026

Corvallis, OR

- Supported operating systems, open-source software, and capstone courses through mentoring, grading, office hours, and technical feedback.
- Guided students on xv6 kernel labs, low-level C, Git collaboration, automated testing, CI, and full-cycle capstone delivery.

Research Assistant

Osborn Popp Lab, Oregon State University

Sept. 2023 - June 2026

Corvallis, OR

- Developed open-source Python tools for NMR spectra visualization, CODEX simulation, curve fitting, and reusable lab data workflows.
- Built NMR Plotter for publication-quality figures and co-authored publications in three peer-reviewed scientific journals.

Web Development Intern

RUSD Solutions

June 2023 - Sept. 2023

Remote

- Contributed to e-commerce deployment, WordPress plugin troubleshooting, and Adobe Acrobat workflow automation.

Projects

SmartTap | *Python, FastAPI, React, TypeScript, Pandas*

2026

- Built a natural-language evidence system for Oregon agriculture and weather data with deterministic validation and CLI, API, and web delivery.

Mocki - BeaverHacks NVIDIA 1st Place | *React, TypeScript, Nemotron*

Apr. 2025

- Built a voice-enabled AI mock interview platform with role-specific interviews, multi-agent orchestration, and structured feedback.

LLM Agents at the Edge - HP Capstone | *Python, AWS, ChromaDB, RAG*

Sept. 2024 - June 2025

- Built an HP-sponsored RAG chatbot for Oregon State Engineering and led its ETL, AWS, and vector-retrieval infrastructure.

NMR Plotter | *Python, Matplotlib, Tkinter*

Jan. 2024 - Apr. 2025

- Developed and published an open-source GUI for NMR visualization and publication-quality figure generation.

Publications

- Metallized, 3D-printed radiofrequency coils.** *Journal of Magnetic Resonance*, 2026. DOI: 10.1016/j.jmr.2026.108057
- NMR_Plotter: A Python GUI for Generating Figures from NMR Spectra.** *Journal of Open Research Software*, 2025. DOI: 10.5334/jors.617
- Measurement of phospholipid lateral diffusion at high pressure by in situ magic-angle spinning NMR spectroscopy.** *Communications Chemistry*, 2025. DOI: 10.1038/s42004-025-01449-7